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Section: **AN41**

MACHINE PROBLEM 1

Screenshots

KNIME DIAGRAM

❖ CSV READER

The screenshot displays the KNIME Analytics Platform interface. The main workspace shows a workflow diagram starting with a 'CSV Reader' node, followed by 'Column Filter', 'Normalizer', and 'Table Partitioner'. The 'Table Partitioner' node is connected to four different machine learning models: 'Decision Tree Learner', 'Logistic Regression Learner', 'Random Forest Learner', and 'Random Forest Predictor'. Each model is followed by a 'Scorer' node. The 'Table Partitioner' node is also connected to a 'Scorer' node. The 'Table Partitioner' node is also connected to a 'Scorer' node. The 'Table Partitioner' node is also connected to a 'Scorer' node.

Below the workflow diagram, the 'Table' view shows the data loaded from the CSV file. The table has 30 rows and 6 columns. The columns are: #, RowID, student_id, attendance, quiz_score, assignment_score, exam_score, and risk_status. The data is as follows:

#	RowID	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk
4	Row3	S004	62	58	60	55	At Risk
5	Row4	S005	55	48	52	50	At Risk

❖ COLUMN FILTER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset

Local - KNIME_project_CS0065_ACT1

Upload

Column Filter

Manual Pattern Type

Search

Excludes

Includes

student_id

attendance

quiz_score

assignment_score

exam_score

risk_status

Any unknown column

Discard Apply Apply and Execute

1: Filtered table

Flow Variables

Rows: 30 Columns: 5 Sorting Filter

Table Statistics

Search

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	95	90	92	88	Not At Risk
2	Row1	88	84	86	82	Not At Risk
3	Row2	76	72	75	70	Not At Risk
4	Row3	62	58	60	55	At Risk
5	Row4	55	48	52	50	At Risk

❖ NORMALIZER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset

Local - KNIME_project_CS0065_ACT1

Upload

Normalizer

Number columns

Manual Pattern Type

Search

Excludes

Includes

No columns in this list.

attendance

quiz_score

assignment_score

exam_score

Any unknown column

Normalization method

Discard Apply Apply and Execute

1: Normalized table 2: Normalize Model

Flow Variables

Rows: 30 Columns: 5 Sorting Filter

Table Statistics

Search

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	0.98	0.92	0.979	0.868	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk
3	Row2	0.592	0.56	0.625	0.528	Not At Risk
4	Row3	0.306	0.28	0.312	0.245	At Risk
5	Row4	0.163	0.08	0.146	0.151	At Risk

❖ TABLE PARTITIONER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader → Column Filter → Normalizer → Table Partitioner

Table Partitioner → Decision Tree Learner → Decision Tree Predictor → Scorer

Table Partitioner → Logistic Regression Learner → Logistic Regression Predictor → Scorer

Table Partitioner → Random Forest Learner → Random Forest Predictor → Scorer

Table Partitioner → Scorer (79%)

Table Partitioner Settings:

- First partition type: Relative (%)
- Relative size: 80
- Sampling strategy: Random
- Group column: risk_status
- Fixed random seed: 42
- If input table is empty: Fail

Table Statistics:

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row1	0.837	0.8	0.854	0.755	Not At Risk
2	Row2	0.592	0.56	0.625	0.528	Not At Risk
3	Row3	0.306	0.28	0.312	0.245	At Risk
4	Row4	0.163	0.08	0.146	0.151	At Risk
5	Row5	0.898	0.86	0.917	0.811	Not At Risk

❖ DECISION TREE LEARNER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader → Column Filter → Normalizer → Table Partitioner

Table Partitioner → Decision Tree Learner → Decision Tree Predictor → Scorer

Table Partitioner → Logistic Regression Learner → Logistic Regression Predictor → Scorer

Table Partitioner → Random Forest Learner → Random Forest Predictor → Scorer

Table Partitioner → Scorer (79%)

Decision Tree Learner Settings:

- Class column: risk_status
- Quality measure: Gini index
- Pruning method: No pruning
- Reduced error pruning: ☒
- Minimum number of records per node: 2
- Number of records to store for view: 10000

The data at the output port is not supported by any modern viewer.

Open port view

❖ DECISION TREE PREDICTOR

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

Decision Tree Learner, Logistic Regression Learner, Random Forest Learner, Decision Tree Predictor, Logistic Regression Predictor, Random Forest Predictor, Scorer

Decision Tree Predictor

Number of patterns for hilling: 10000

☐ Custom prediction column name

☐ Append columns with normalized class distribution

Discard Apply Apply and Execute

► 1: Classified Data

Rows: 6 | Columns: 6

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_status)
1	Row0	0.98	0.92	0.979	0.868	Not At Risk	Not At Risk
2	Row6	0.429	0.42	0.333	0.34	At Risk	At Risk
3	Row7	0.714	0.68	0.729	0.642	Not At Risk	Not At Risk
4	Row25	0.286	0.26	0.292	0.226	At Risk	At Risk
5	Row28	0.755	0.72	0.771	0.698	Not At Risk	Not At Risk

❖ SCORER 1

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

Decision Tree Learner, Logistic Regression Learner, Random Forest Learner, Decision Tree Predictor, Logistic Regression Predictor, Random Forest Predictor, Scorer

Scorer

First column: Prediction (risk_status)

Second column: risk_status

Sorting strategy: Insertion order

☐ Reverse order

☐ Use name prefix

Missing values: Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix

Rows: 2 | Columns: 2

#	RowID	Not At Risk	At Risk
1	Not A...	3	0
2	At Risk	0	3

❖ LOGISTIC REGRESSION LEARNER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1 +

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer K-ANALYTICS Monitor

1: Model for Predictor 2: Coefficients and Statistics 3: Model and Learning Properties Flow Variables

The data at the output port is not supported by any modern viewer.

Open port view

Logistic Regression Learner

Target column: risk_status

Reference category: At Risk

☐ Use order from target column domain

Solver: Stochastic average gradient

Feature selection: Manual Pattern Type

Excludes: Includes: attendance

Discard Apply Apply and Execute

❖ LOGISTIC REGRESSION PREDICTOR

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1 +

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer K-ANALYTICS Monitor

1: Predicted data Flow Variables

Rows: 6 | Columns: 6 Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_status)
1	Row0	0.98	0.92	0.979	0.868	Not At Risk	Not At Risk
2	Row6	0.429	0.42	0.333	0.34	At Risk	Not At Risk
3	Row7	0.714	0.68	0.729	0.642	Not At Risk	Not At Risk
4	Row25	0.286	0.26	0.292	0.226	At Risk	At Risk
5	Row28	0.755	0.72	0.771	0.698	Not At Risk	Not At Risk

Logistic Regression Predictor

☐ Custom prediction column name

☐ Append columns with normalized class distribution

Discard Apply Apply and Execute

❖ SCORER 2

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Logistic Regression Learner Random Forest Learner

Decision Tree Predictor Logistic Regression Predictor Random Forest Predictor

Scorer

First column: T risk_status

Second column: T risk_status

Same column selected. The confusion matrix requires two distinct columns.

Sorting strategy: Insertion order

Reverse order: ☐ Use name prefix: ☐

Missing values: Discard

Apply Apply and Execute

1: Confusion matrix 2: Accuracy statistics Flow Variables

Rows: 2 Columns: 2 Sorting Filter

#	RowID	Not At Risk	At Risk
1	Not A...	3	0
2	At Risk	0	3

❖ RANDOM FOREST LEARNER

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Logistic Regression Learner Random Forest Learner

Decision Tree Predictor Logistic Regression Predictor Random Forest Predictor

Scorer

Random Forest Learner

Attribute Selection

Target column: T risk_status

Training attributes: Use column Use fingerprint

Attribute selection: Manual Pattern

Excludes: Includes: attendance quiz_score assignment_score

Discard Apply Apply and Execute

1: Out-of-bag Predictions 2: Attribute Statistics 3: Random Forest Model Flow Variables

Rows: 24 Columns: 10 Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	P (risk_status=N...	P (risk_status=A...	risk_status (Out-...	risk_status (Out-...	model count
1	Row1	0.837	0.8	0.854	0.755	Not At Risk	0	0	0	0	0
2	Row2	0.592	0.56	0.625	0.528	Not At Risk	0	0	0	0	0
3	Row3	0.306	0.28	0.312	0.245	At Risk	0	0	0	0	0
4	Row4	0.163	0.08	0.146	0.151	At Risk	0	0	0	0	0
5	Row5	0.898	0.86	0.917	0.811	Not At Risk	0	0	0	0	0

RANDOM FOREST PREDICTOR

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1 +

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Logistic Regression Learner Random Forest Learner

Decision Tree Predictor Logistic Regression Predictor Random Forest Predictor

Scorer Scorer Scorer

Random Forest Predictor

Change prediction column name

Prediction column name

Prediction (exam_score)

☒ Append overall prediction confidence

☐ Append individual class probabilities

Suffix for probability columns

☐ Use soft voting

Discard Apply Apply and Execute

1: Prediction output Flow Variables

Rows: 6 | Columns: 7

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)	Prediction (risk_status) (# String)	Prediction (risk_status) (...) (# Number (Float))	
☐	1	Row0	0.98	0.92	0.979	0.868	Not At Risk	Not At Risk	1
☐	2	Row6	0.429	0.42	0.333	0.34	At Risk	At Risk	1
☐	3	Row7	0.714	0.68	0.729	0.642	Not At Risk	Not At Risk	1
☐	4	Row25	0.286	0.26	0.292	0.226	At Risk	At Risk	1
☐	5	Row28	0.755	0.72	0.771	0.698	Not At Risk	Not At Risk	1

SCORER 3

KNIME Analytics Platform

Home KNIME_project_CS0065_ACT1 +

Save Execute Cancel Reset Local - KNIME_project_CS0065_ACT1 Upload

Info Nodes Explorer KAI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Logistic Regression Learner Random Forest Learner

Decision Tree Predictor Logistic Regression Predictor Random Forest Predictor

Scorer Scorer Scorer

Scorer

First column

Prediction (risk_status)

Second column

risk_status

Sorting strategy

Insertion order

☐ Reverse order

☐ Use name prefix

Missing values

Ignore Fail

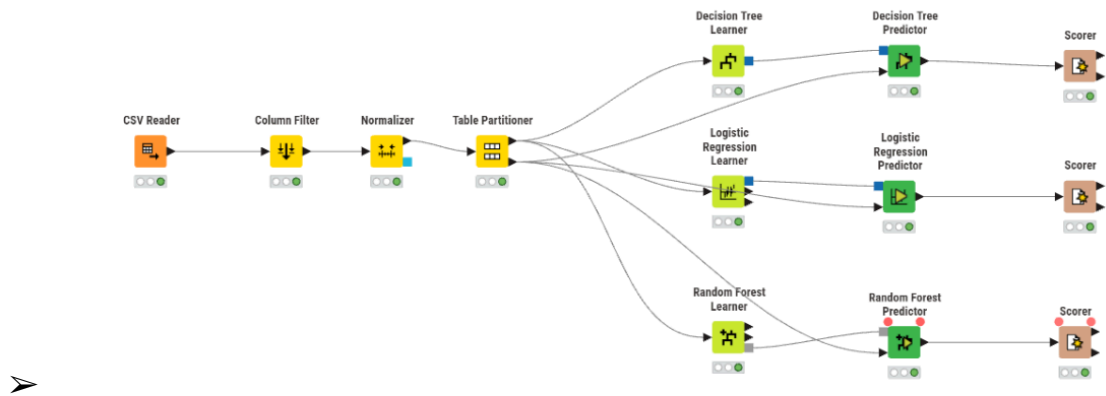
Discard Apply Apply and Execute

1: Confusion matrix 2: Accuracy statistics Flow Variables

Rows: 2 | Columns: 2

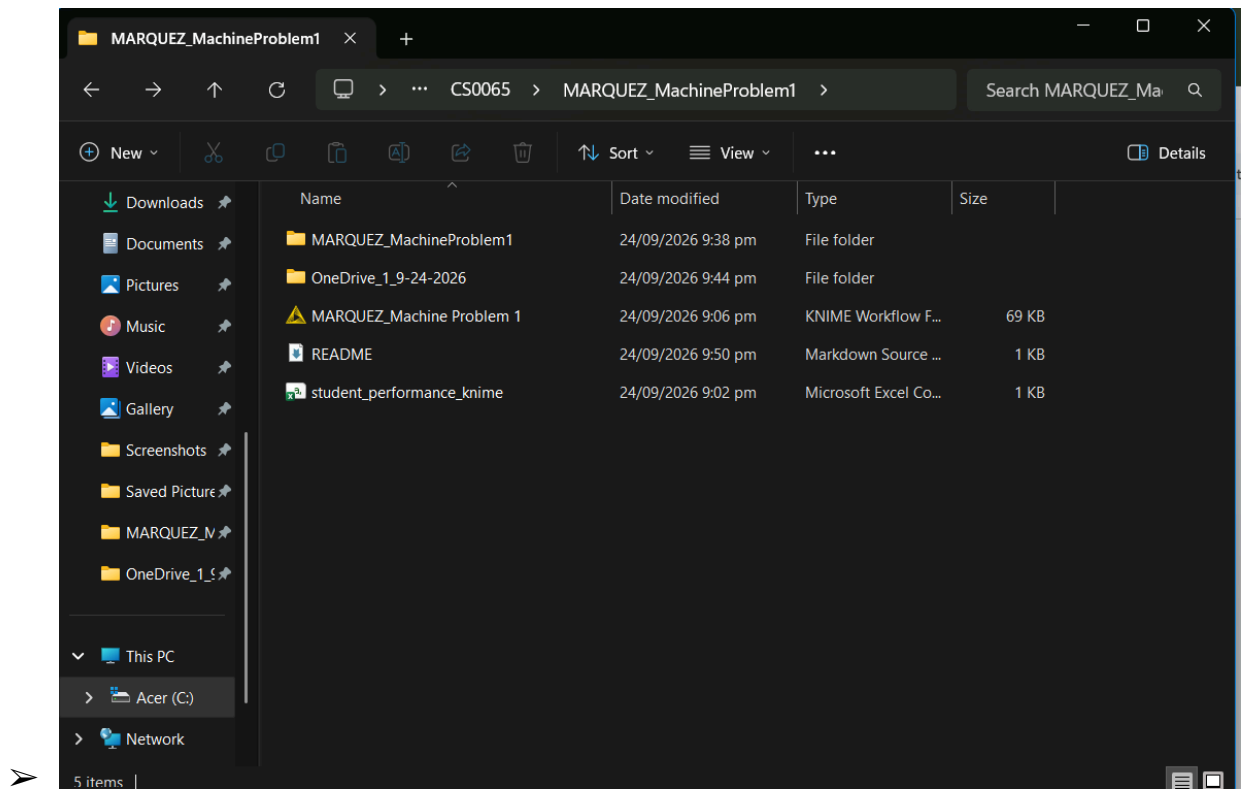
#	RowID	Not At Risk (# Number (Integer))	At Risk (# Number (Integer))
☐	1	Not A...	3
☐	2	At Risk	0

❖ DIAGRAM

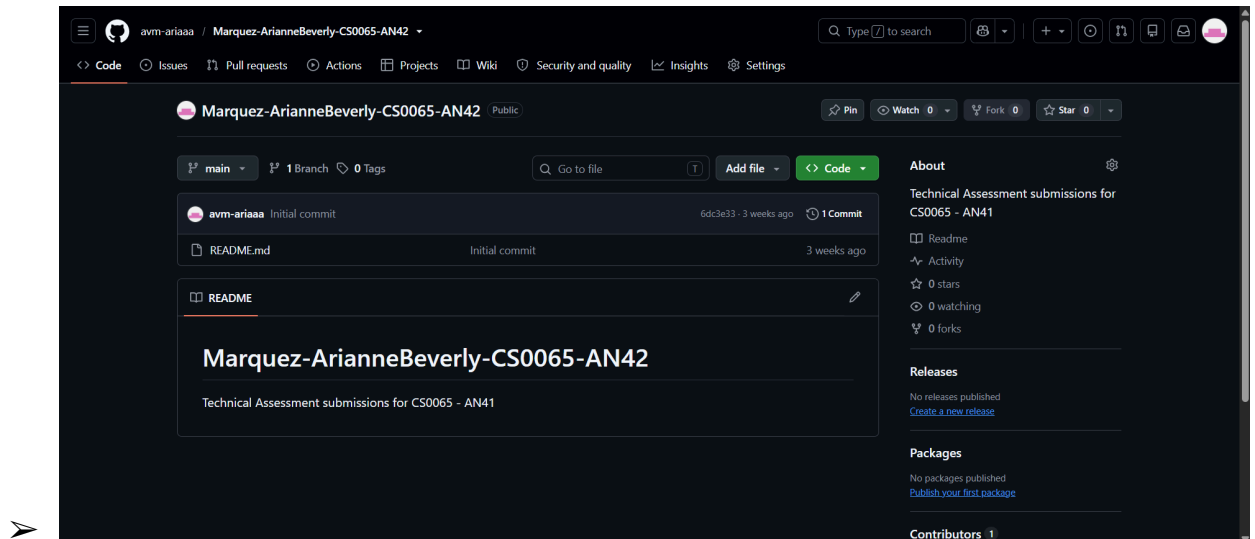


FILE ORGANIZATION (FROM DEVICE DIRECTORY TO GITHUB REPOSITORY)

- ❖ **MARQUEZ_MachineProblem1** folder where all the required files are restored upon requirements



- ❖ The GitHub Repository where the activity will be stored (Note: there was a typo wherein “AN42” was the section, it was later changed to “AN41” which is the proper section)



- ❖ Initializing Git Bash to upload the necessary files

```
MINGW64:/c/Knime/CS0065/MARQUEZ_MachineProblem1
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1
$ git init
Initialized empty Git repository in C:/Knime/CS0065/MARQUEZ_MachineProblem1/.git/
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (master)
$ git branch -M main
```

```
MINGW64:/c/Knime/CS0065/MARQUEZ_MachineProblem1
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main)
$ git remote add origin https://github.com/avm-ariaaa/Marquez-ArianneBeverly-CS0065-AN41.git
```

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main)
$ git pull origin main --allow-unrelated-histories --no-rebase --no-edit
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 1.87 KiB | 191.00 KiB/s, done.
From https://github.com/avm-ariaaa/Marquez-ArianneBeverly-CS0065-AN41
* branch          main          -> FETCH_HEAD
* [new branch]     main          -> origin/main
Auto-merging README.md
CONFLICT (add/add): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

➤

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main|MERGING)
$ git add .
```

➤

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main|MERGING)
$ git commit -m "Add Machine Problem 1 files"
[main 07c823b] Add Machine Problem 1 files
```

➤

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 147, done.
Counting objects: 100% (147/147), done.
Delta compression using up to 12 threads
Compressing objects: 100% (131/131), done.
Writing objects: 100% (145/145), 345.70 KiB | 6.91 MiB/s, done.
Total 145 (delta 46), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (46/46), done.
To https://github.com/avm-ariaaa/Marquez-ArianneBeverly-CS0065-AN41.git
   e061216..07c823b  main -> main
branch 'main' set up to track 'origin/main'.
```

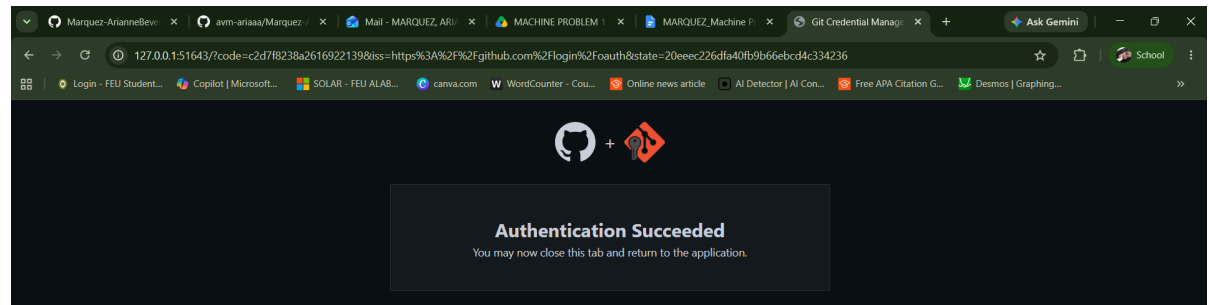
➤

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main)
$ git commit --amend -m "Machine Problem 1 files"
[main 6b4ca9d] Machine Problem 1 files
Date: Thu Sep 24 23:07:10 2026 +0800
```

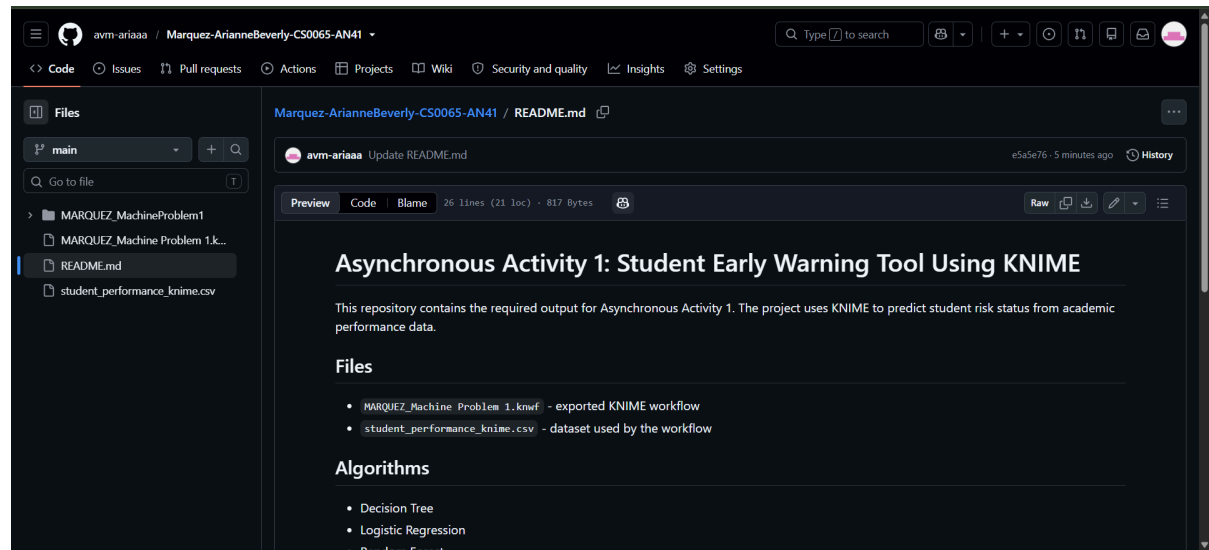
➤

```
Yani@LAPTOP-J91IDR71 MINGW64 /c/Knime/CS0065/MARQUEZ_MachineProblem1 (main)
$ git push --force
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 486 bytes | 486.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/avm-ariaaa/Marquez-ArianneBeverly-CS0065-AN41.git
+ 07c823b...6b4ca9d main -> main (forced update)
```

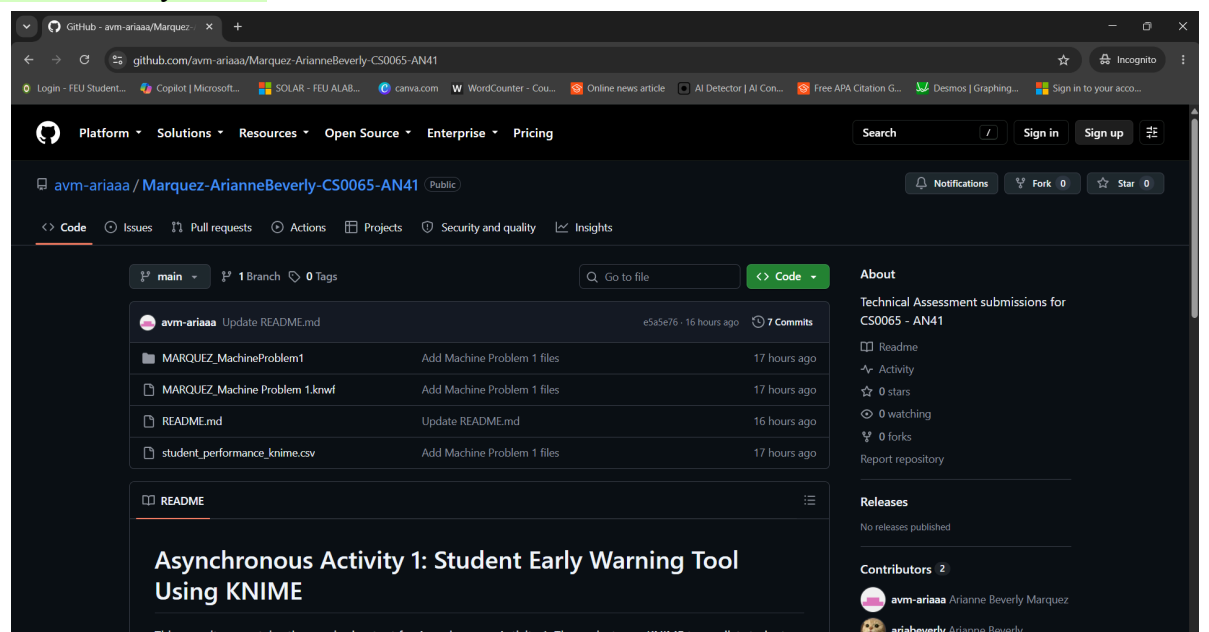
❖ Git Authentication



❖ Git Repository



❖ Project Accessibility Check



Note:

The process of adding this pdf file was not added in the screenshots as that was the last step done for this activity. This file was saved as pdf first before it was added in the folder, and uploaded in Github. Rest assured that it was done using the same process, with the same syntaxes used for uploading the files for this activity in Github via Gitbash.